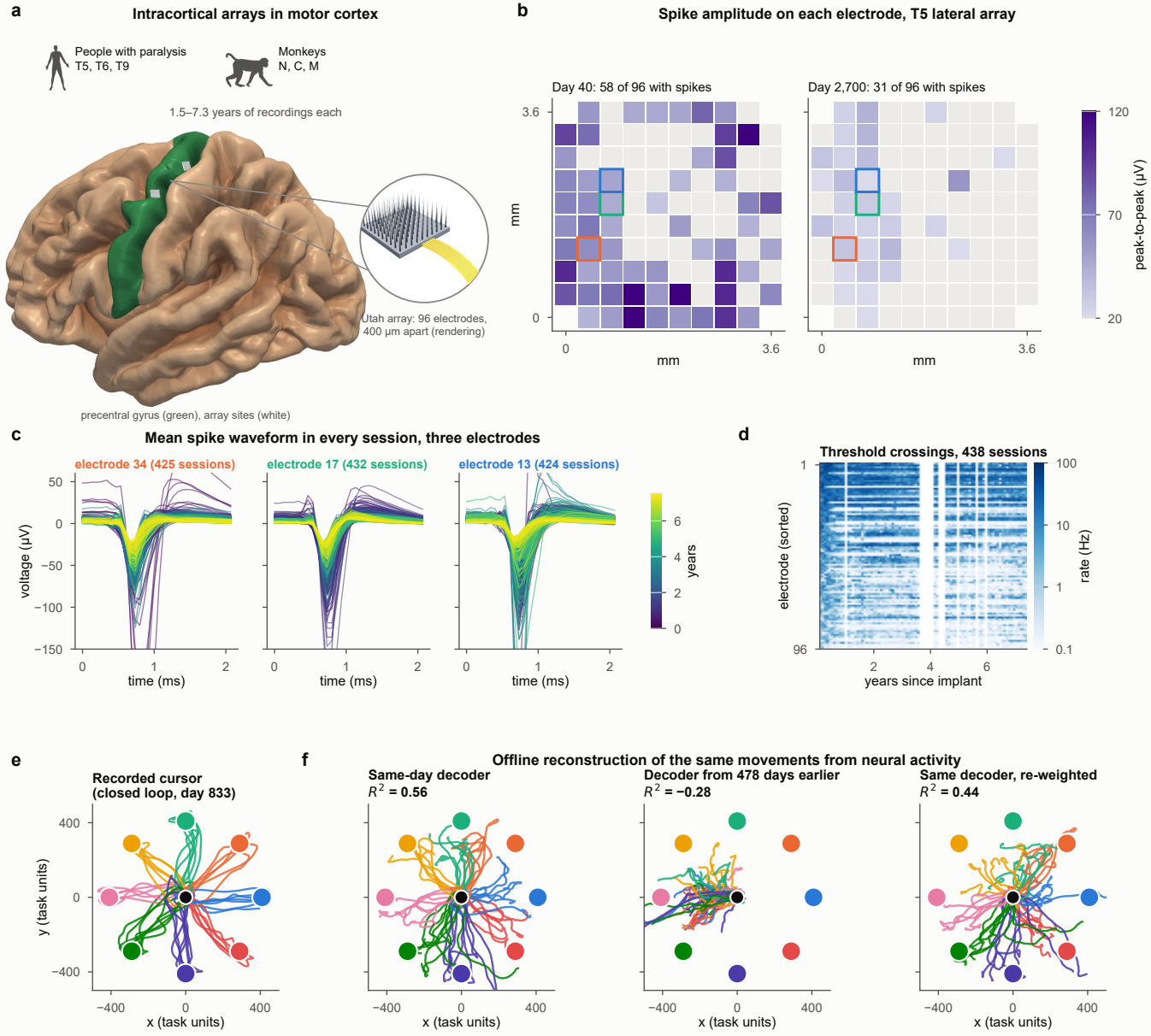

a
Intracortical arrays in motor cortex
People with paralysis
T5, T6, T9
Monkeys
N, C, M
1.5–7.3 years of recordings each
Utah array: 96 electrodes,
400 µm apart (rendering)
precentral gyrus (green), array sites (white)
b
Spike amplitude on each electrode, T5 lateral array
Day 40: 58 of 96 with spikes
Day 2,700: 31 of 96 with spikes
3.6
0
mm
peak-to-peak (µV)
120
70
20
c
Mean spike waveform in every session, three electrodes
electrode 34 (425 sessions)
electrode 17 (432 sessions)
electrode 13 (424 sessions)
voltage (µV)
time (ms)
years
d
Threshold crossings, 438 sessions
electrode (sorted)
years since implant
rate (Hz)
e
Recorded cursor
(closed loop, day 833)
y (task units)
x (task units)
f
Offline reconstruction of the same movements from neural activity
Same-day decoder
R^2 = 0.56
Decoder from 478 days earlier
R^2 = −0.28
Same decoder, re-weighted
R^2 = 0.44
x (task units)